**Micro I, class 3**

I started by writing down the *Kuhn-Tucker conditions* for the (UM) problem (reported on in my *Concave Programming* notes). I pointed out that they could in principle be used for two things: to find functional forms for estimation or calibration; and to derive testable implications of consumer theory. After some reflection we thought better of these applications of the conditions in the context of demand theory (though they are are still useful for other things).

Instead I considered the *indirect utility function*, the value function $V(p, w)$ of problem (UM). Assuming continuous differentiability of the representation $u$, the demand $d$ and the multiplier $\lambda^*$ (the last two viewed as functions of $(p, w)$), we agreed that

$$\frac{\partial V}{\partial w}\big(p, w\big) = \lambda^*(p, w).$$

This is a special case of (the classical) Envelope Theorem to which we next turn. It implies *Roy's Identity*, expressing demand as the ratio of two indirect-utility-function derivatives,

$$d_i(p, w) = -\frac{\partial V/\partial p_i}{\partial V/\partial w}\big(p, w\big),$$

provided of course that $\partial V/\partial w > 0$. As mentioned, this identity gives a generally-much-simpler way to calculate demands than the Kuhn-Tucker conditions—provided that we know an indirect utility when we see one, which will be a main event the next class. I concluded with a brief discussion of orange juice and milk,